

The three-sparse Walsh threshold class: an exact count and a one-bit VC-dimension gap

New partial result for Problem 8 of arXiv:1507.04319

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Status. **New partial result; the arbitrary-sparsity problem remains open.** For one and two terms the VC dimension is exactly $n + 1$. For three terms the classifiers admit a complete algebraic description and exact count. This gives exact answers in dimensions $n \leq 3$, and for every $n \geq 4$ it determines the VC dimension up to one.

Source question

Mixon and Peterson [1] define $C_{n,k}$ to be the functions $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$ of the form

$$g(x) = \text{sign} \left(\sum_{i=1}^k a_i \prod_{j \in S_i} x_j \right)$$

and ask in Problem 8: “Determine the VC dimension of $C_{n,k}$.” They prove $\max\{n, k\} \leq \text{VCdim}(C_{n,k}) \leq 2nk - O(k \log k)$.

6. CONCLUSION AND OPEN PROBLEMS

This paper demonstrates the plausibility of learning Boolean functions with concentrated spectra, as well as its applicability to modern classification theory and application. However, this paper offers more questions than answers. For example, while we showed in Section 2 that the VC dimension h of $C_{n,k}$ satisfies

$$\max\{n, k\} \leq h \leq 2nk - O(k \log k),$$

we have yet to identify how h scales with n and k . This leads to our first open problem:

Problem 8. Determine the VC dimension of $C_{n,k}$.

In Section 3, we proposed an algorithm for empirical risk minimization over polynomials in $C_{n,k}$ of degree at most d . However, we still don’t know if this algorithm produces an estimate $\hat{f}$ that satisfies a guarantee of the form [\(5\)](#).

Source evidence: arXiv:1507.04319, p. 7, Problem 8.

We identify the cube with $V = \mathbb{F}_2^n$, write $\chi_a(x) = (-1)^{a \cdot x}$, and encode a sign function as $g(x) = (-1)^{f(x)}$ with $f : V \rightarrow \mathbb{F}_2$. An affine Boolean form is a function $\ell(x) = b + a \cdot x$.

Proof intuition

Two Walsh characters are rigid: after factoring one character, the sign of $a + b\chi$ is simply the sign of its larger coefficient, unless the expression vanishes. Three characters are the first genuinely new case. After factoring one character they give a linear threshold function on a two-dimensional Boolean square. The only non-affine possibilities have one exceptional vertex. In algebraic normal form these are exactly $\ell + L_1 L_2$.

The quadratic part of $L_1 L_2$ is the decomposable bivector $u \wedge v$, so the nonzero quadratic parts are indexed by two-planes in $\mathbb{F}_2^n$. This gives the exact count and hence the upper bound. For the lower bound, start with $0, e_1, \dots, e_n$, then add $e_i + e_j$ for every edge of $K_{2,n-2}$. Affine terms fit arbitrary labels on the affine basis. Each right vertex of $K_{2,n-2}$ then stores its two remaining second differences as one vector in $\mathbb{F}_2^2$, making all edge labels independent. The source problem for larger k , and one bit for $k = 3, n \geq 4$, are not settled by this argument.

Lemma 1 (two-term rigidity). *For every $n \geq 1$,*

$$C_{n,2} = C_{n,1} = \{\pm \chi_a : a \in \mathbb{F}_2^n\}, \quad \text{VCdim}(C_{n,1}) = \text{VCdim}(C_{n,2}) = n + 1.$$

Proof. After combining repeated characters, a two-term expression with distinct characters factors as

$$a\chi_r + b\chi_s = \chi_r(a + b\chi_{r+s}).$$

If its sign is defined everywhere, then $|a| \neq |b|$. If $|a| > |b|$, its sign is $\text{sign}(a)\chi_r$; if $|b| > |a|$, its sign is $\text{sign}(b)\chi_s$. Thus two terms give no new classifiers.

The 2^{n+1} functions $\pm \chi_a$ are precisely the phases of affine Boolean forms. They shatter $\{0, e_1, \dots, e_n\}$, since an affine form can be prescribed arbitrarily on an affine basis. The class-size bound gives the reverse inequality. $\square$

Proposition 2 (classification and exact count). *The phases in $C_{n,3}$ are exactly*

$$(-1)^{\ell(x) + L_1(x)L_2(x)}, \tag{1}$$

where $\ell, L_1, L_2 : V \rightarrow \mathbb{F}_2$ are affine forms. Consequently

$$|C_{n,3}| = 2^{n+1} \left(1 + \begin{bmatrix} n \\ 2 \end{bmatrix}_2 \right) = 2^{n+1} \left(1 + \frac{(2^n - 1)(2^n - 2)}{6} \right), \tag{2}$$

where the Gaussian coefficient is understood as zero for $n < 2$.

Proof. Consider three distinct characters and factor the first:

$$\text{sign}(a\chi_r + b\chi_s + c\chi_t) = \chi_r \text{sign}(a + b\chi_{r+s} + c\chi_{r+t}).$$

The nonzero vectors $r + s$ and $r + t$ are distinct and hence linearly independent over $\mathbb{F}_2$. The last sign is an affine real threshold dichotomy on the four points of $\{\pm 1\}^2$. Of the sixteen dichotomies, parity and its complement are the only two that are not linearly separable. The remaining fourteen are constants, signed coordinates, or dichotomies with exactly one exceptional vertex. Constants and coordinates give affine Boolean phases. An exceptional-vertex dichotomy has the form $(-1)^{L_1 L_2}$, up to an affine phase. Conversely,

$$(-1)^{pq} = \text{sign}(1 + (-1)^p + (-1)^q) \quad (p, q \in \mathbb{F}_2),$$

so every phase in (1) is the sign of at most three characters.

Write the linear parts of L_1, L_2 as $u \cdot x, v \cdot x$. The quadratic part of $L_1 L_2$ is

$$\sum_{i < j} (u_i v_j + u_j v_i) x_i x_j,$$

the bivector $u \wedge v$. It is zero when u, v are dependent. If it is nonzero, it uniquely determines the plane $\langle u, v \rangle$: over $\mathbb{F}_2$, every ordered basis of that plane has determinant one and hence the same nonzero wedge. Thus the possible quadratic parts are zero and one for each two-plane of V^* . For each quadratic part, the unique algebraic normal form allows exactly 2^{n+1} affine perturbations. The number of two-planes is $\begin{bmatrix} n \\ 2 \end{bmatrix}_2 = (2^n - 1)(2^n - 2)/6$, proving (2). $\square$

Theorem 3 (one-bit determination for three terms). *The three-sparse class satisfies*

$$\text{VCdim}(C_{1,3}) = 2, \quad \text{VCdim}(C_{2,3}) = 4, \quad \text{VCdim}(C_{3,3}) = 7.$$

For every $n \geq 4$,

$$3n - 3 \leq \text{VCdim}(C_{n,3}) \leq 3n - 2. \quad (3)$$

Proof. We first prove the lower bound for $n \geq 4$. Let

$$S = \{0, e_1, \dots, e_n\} \cup \{e_i + e_j : i \in \{1, 2\}, 3 \leq j \leq n\}.$$

Thus $|S| = n + 1 + 2(n - 2) = 3n - 3$. Prescribe arbitrary labels $y : S \rightarrow \mathbb{F}_2$. For $i \in \{1, 2\}$, $j \geq 3$, set the requested second difference

$$d_{ij} = y(0) + y(e_i) + y(e_j) + y(e_i + e_j).$$

Choose $u, v \in \mathbb{F}_2^n$ by

$$(u_1, v_1) = (1, 0), \quad (u_2, v_2) = (0, 1), \quad (u_j, v_j) = (d_{2j}, d_{1j}) \quad (j \geq 3).$$

For $Q(x) = (u \cdot x)(v \cdot x)$, the coefficient of $x_i x_j$ is $u_i v_j + u_j v_i = d_{ij}$ on every displayed edge. Choose the unique affine form ℓ such that $\ell + Q$ agrees with y on $0, e_1, \dots, e_n$. The four-point affine cancellation identity then gives, for every added pair-sum point,

$$(\ell + Q)(0) + (\ell + Q)(e_i) + (\ell + Q)(e_j) + (\ell + Q)(e_i + e_j) = d_{ij},$$

so $\ell + Q = y$ throughout S . Proposition 2 shows that this is a member of $C_{n,3}$. Hence S is shattered.

For $n = 3$, take $S = \{x \in \mathbb{F}_2^3 : |x| \leq 2\}$. Every alternating quadratic form in three variables has rank at most two and is therefore decomposable. The same affine-interpolation argument realizes all labels on these seven points. Formula (2) gives $|C_{3,3}| = 128$, so the VC dimension is seven. For $n = 2$, every Boolean quadratic has decomposable quadratic part, so Proposition 2 gives all 16 functions; for $n = 1$, it gives all four.

Finally, for $n \geq 4$, (2) implies

$$|C_{n,3}| = 2^{n+1} \left(1 + \frac{(2^n - 1)(2^n - 2)}{6} \right) < 2^{3n-1}.$$

A class shattering $3n - 1$ points would contain at least 2^{3n-1} distinct functions. Therefore $\text{VCdim}(C_{n,3}) \leq 3n - 2$, completing (3). $\square$

Verification, limitations, and upgrade audit

The proof is symbolic. The bundled script independently checks $C_{n,2} = C_{n,1}$ for $1 \leq n \leq 5$, the exact count for $1 \leq n \leq 5$, and every target labeling produced by the explicit shattering synthesis for $3 \leq n \leq 6$. It also exhaustively computes $\text{VCdim}(C_{n,3}) = 2, 4, 7, 9$ for $n = 1, 2, 3, 4$, consistent with the theorem. These finite checks are sanity checks, not substitutes for the proof.

The remaining one-bit upgrade has a precise exterior-algebra formulation. After quotienting restrictions by affine functions, shattering $3n - 2$ points would require a $2n - 3$ -dimensional point-evaluation quotient of $\bigwedge^2 \mathbb{F}_2^n$ in which every vector has a decomposable representative. The construction above attains dimension $2n - 4$. Direct quotient searches for $n = 5, 6$ found no final-dimension example, but no proof excluding one was found; no exact claim is made for $n \geq 4$.

Exact-title, exact-question, Walsh–Hadamard VC-dimension, sparse polynomial threshold, and threshold-density searches through 13 August 2026 found no later source settling this exact problem or the formulas above. The search was bounded rather than a systematic bibliographic review. Human review should focus on the two-variable threshold classification, uniqueness in the exact count, and the $K_{2,n-2}$ second-difference synthesis. The original problem for arbitrary k remains open.

Reference

[1] D. G. Mixon and J. Peterson, *Learning Boolean functions with concentrated spectra*, arXiv:1507.04319 (2015), especially Sections 2 and 6, Problem 8.